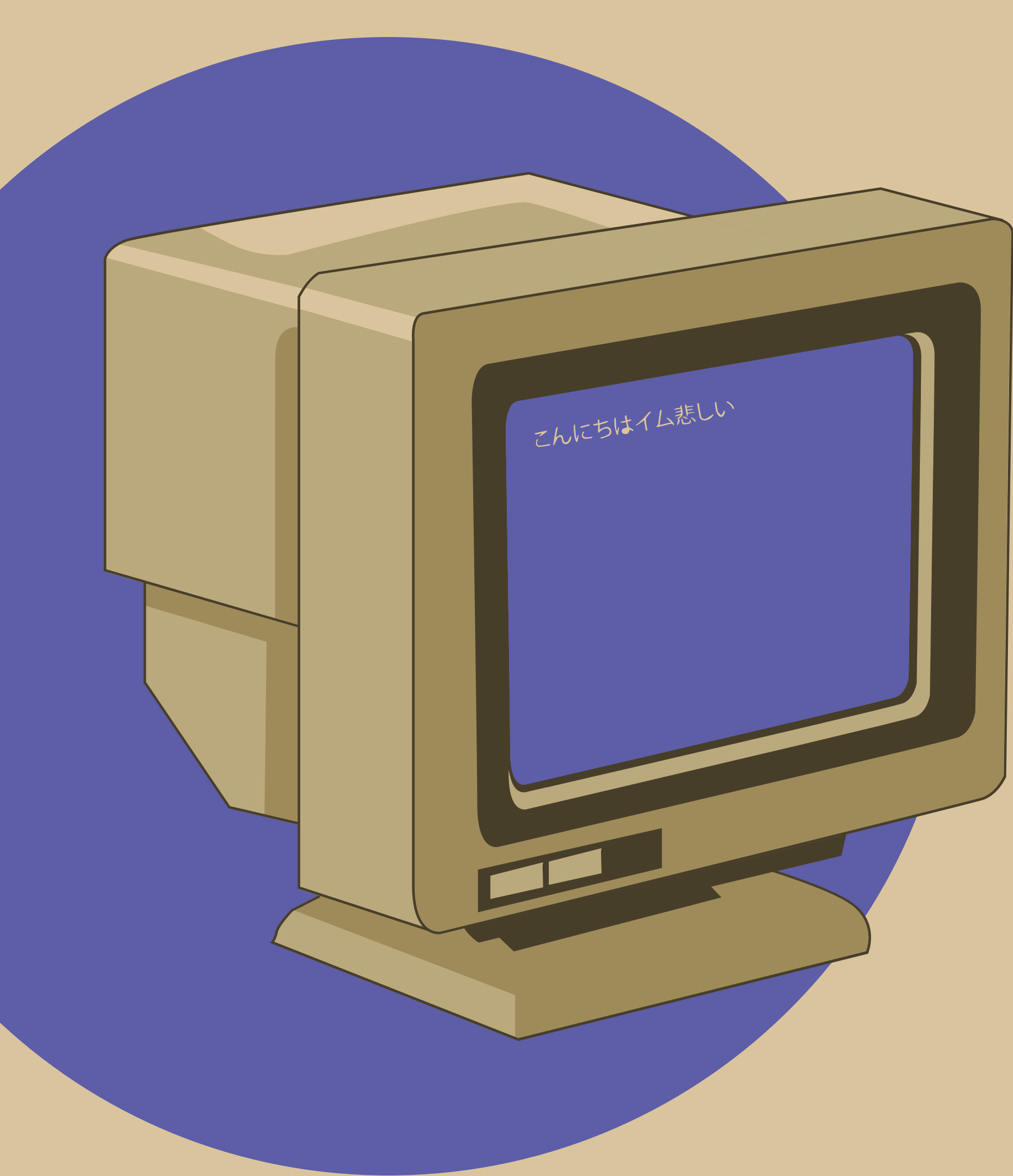




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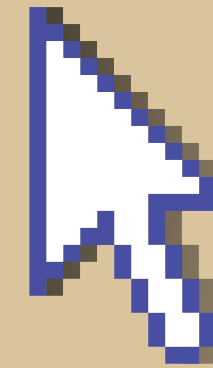


STUXNET

COMPUTER WORM

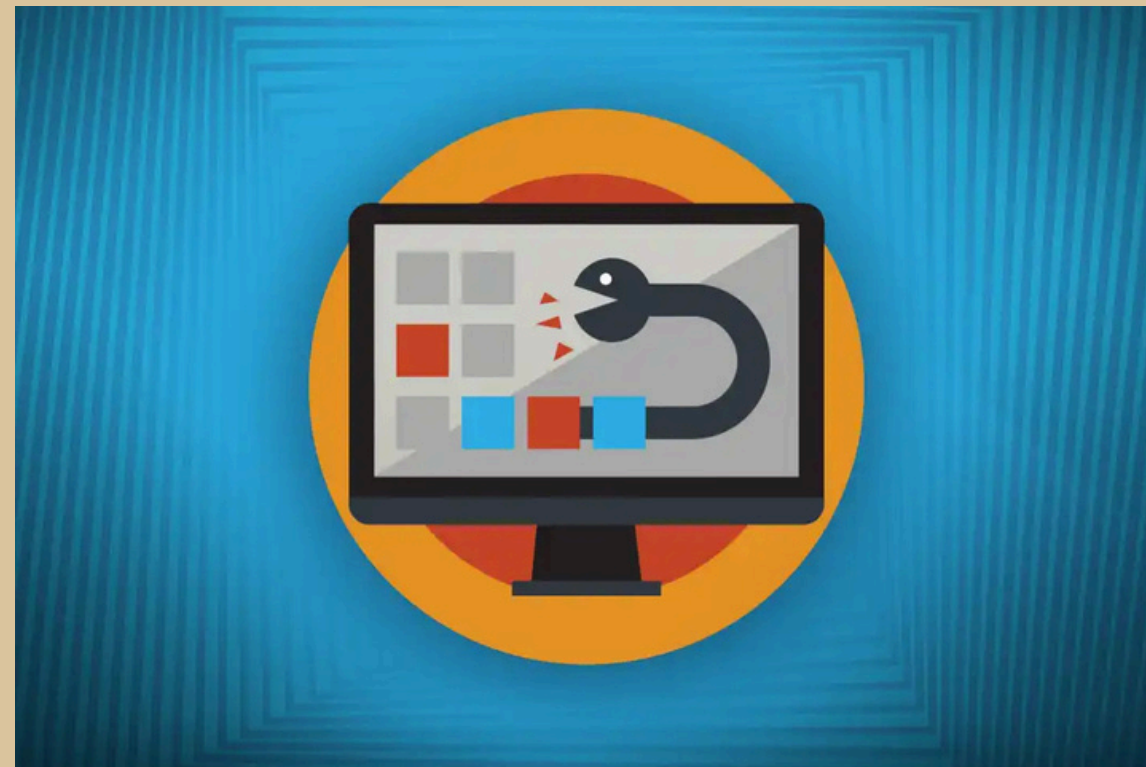


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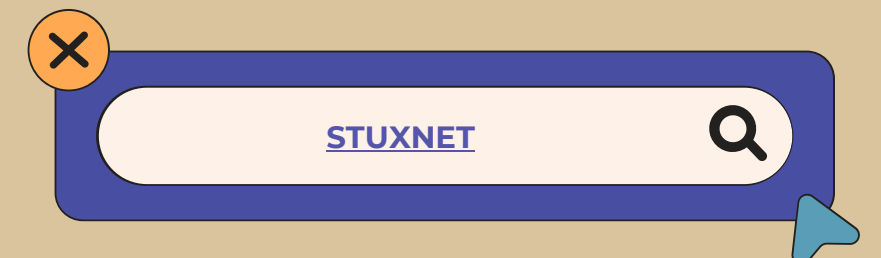
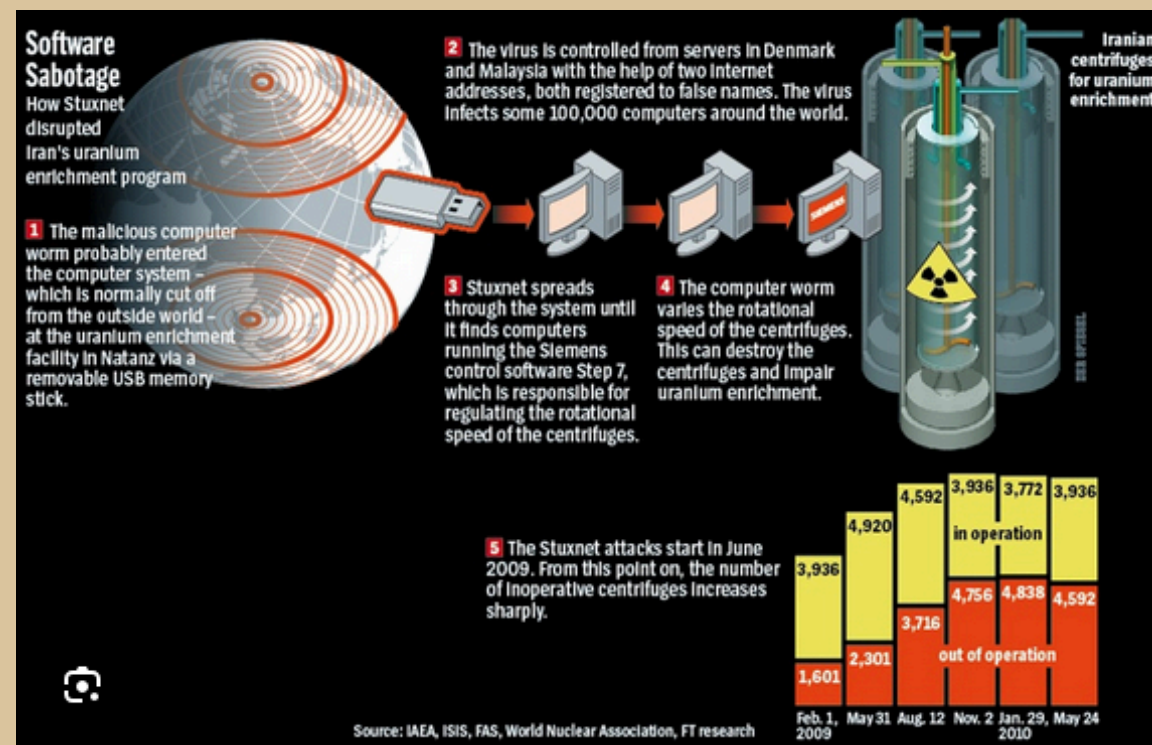


WHAT IS STUXNET WORM?

Stuxnet is a computer worm discovered in June 2010 that targeted Iran's Natanz nuclear facility.

It was the first known cyber weapon to cause real-world physical damage.

Unlike normal malware that steals or corrupts data, Stuxnet was created to sabotage industrial machines.





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ESTIMATED DATE AND WHEREABOUT

WHEN ?

Active around 2009–2010,
discovered mid-2010.



WHERE ?

Mainly in Iran, but it spread
globally across computers
running Windows.



DISCOVER ?

First found by a Belarus
security company,
VirusBlokAda, after unusual
system crashes in Iran.



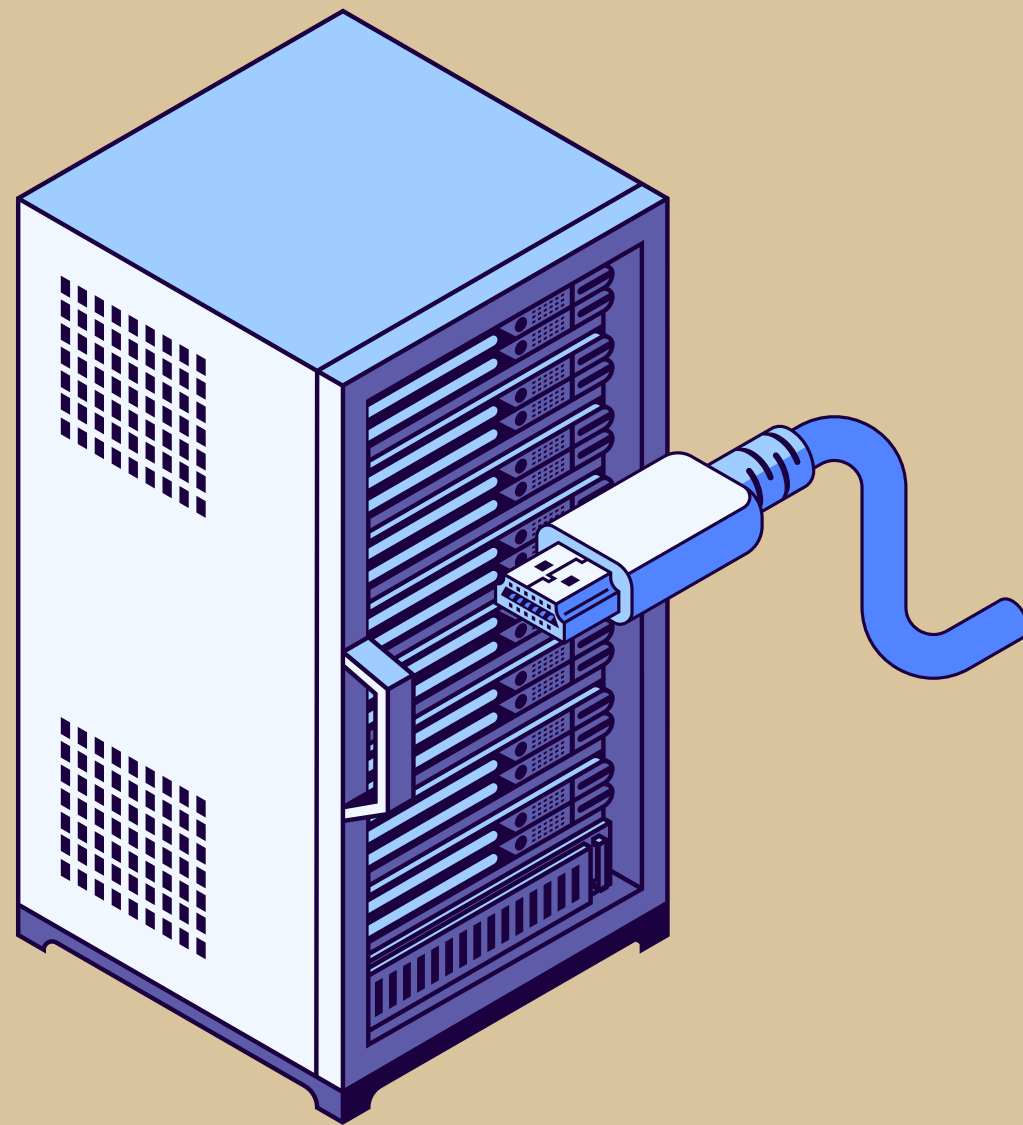
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MAIN TARGET



1.

Target System

Siemens Step7 software controlling Programmable Logic Controllers (PLCs).

2.

Goal

Disrupt uranium enrichment by damaging centrifuges that separate nuclear material.

3.

Victim Site

Iran's Natanz nuclear facility.



STUXNET

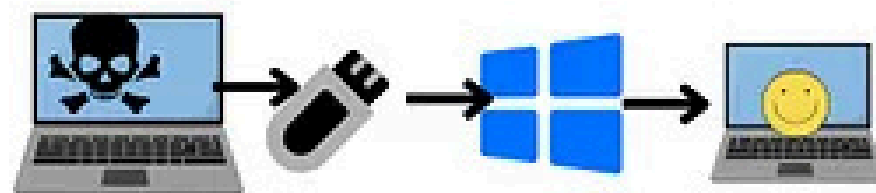




PROCEDURES



HOW STUXNET WORKED



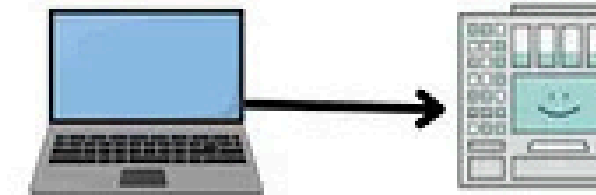
1. infection

Stuxnet enters a system via a USB stick and proceeds to infect all machines running Microsoft Windows. By brandishing a digital certificate that seems to show that it comes from a reliable company, the worm is able to evade automated-detection systems.



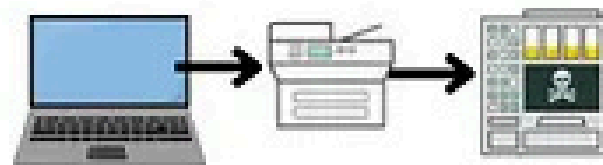
2. search

Stuxnet then checks whether a given machine is part of the targeted industry- trial control system made by Siemens. Such systems are deployed in Iran to run high-speed centrifuges that help to enrich nuclear fuel.



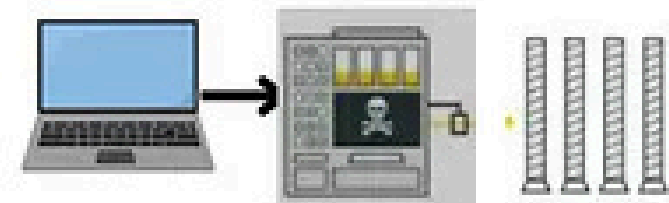
3. update

If the system isn't a target, Stuxnet does nothing; if it is, the worm attempts to access the Internet and download a more recent version of itself



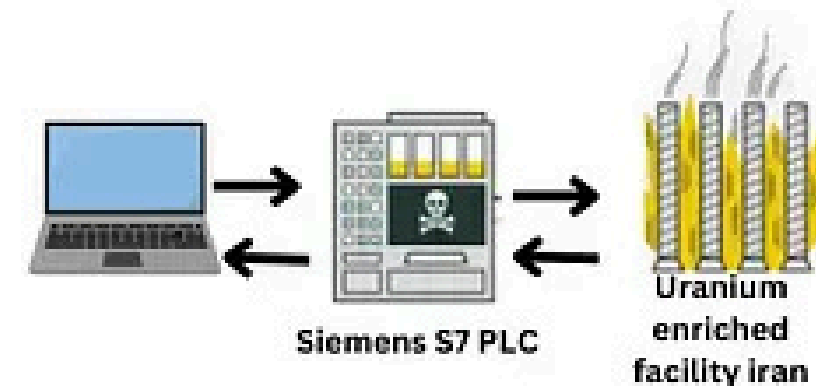
4. compromise

The worm then compromises the target system's logic controllers, exploiting "zero day" vulnerabilities- software weaknesses that haven't been identified by security experts.



5. control

In the beginning, Stuxnet spies on the operations of the targeted system. Then it uses the information it has gathered to take control of the centrifuges, making them spin themselves to failure.



6. deceive and destroy

Meanwhile, it provides false feed- back to outside controllers, ensuring that they won't know what's going wrong until it's too late to do anything about it

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1. Infection
2. SEARCH
3. UPDATE
4. COMPROMISE
5. CONTROL
6. DECEIVE AND DESTROY



MEDIUM

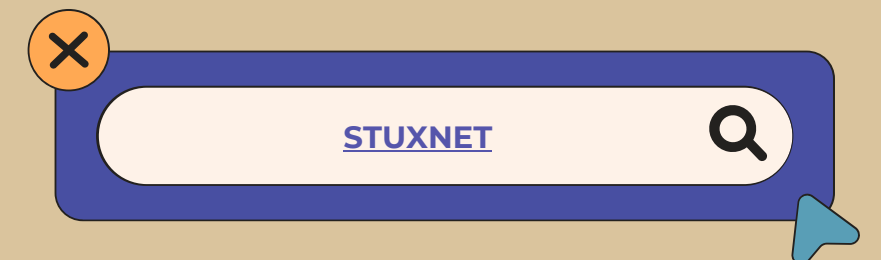




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TECHNICAL HIGHLIGHTS

- Used four Windows zero-day vulnerabilities (previously unknown flaws).
- Used stolen digital certificates from Realtek & JMicron to appear legitimate.
- Spread via USB drives to jump into isolated (air-gapped) networks.
- Once inside, it reprogrammed PLCs to spin centrifuges too fast or too slow.
- Displayed fake readings so operators saw nothing wrong.





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IMPACT OF THE ATTACK

- Destroyed or damaged over **1,000 centrifuges**.
- **Delayed Iran's nuclear enrichment** progress by months or years.
- Proved that cyber attacks can cause **physical destruction**.
- **Inspired** other attacks on industrial control systems (ICS) worldwide.



STUXNET





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WHO CREATED IT?

- No official confirmation, but evidence points to a U.S.–Israel operation named “**Operation Olympic Games.**”
- Goal: **Slow down Iran’s nuclear development** without direct military conflict.



STUXNET

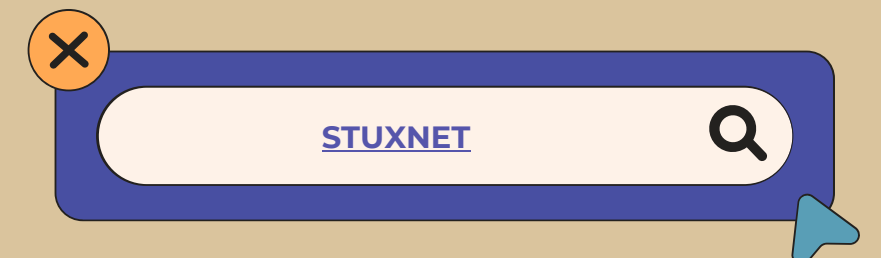
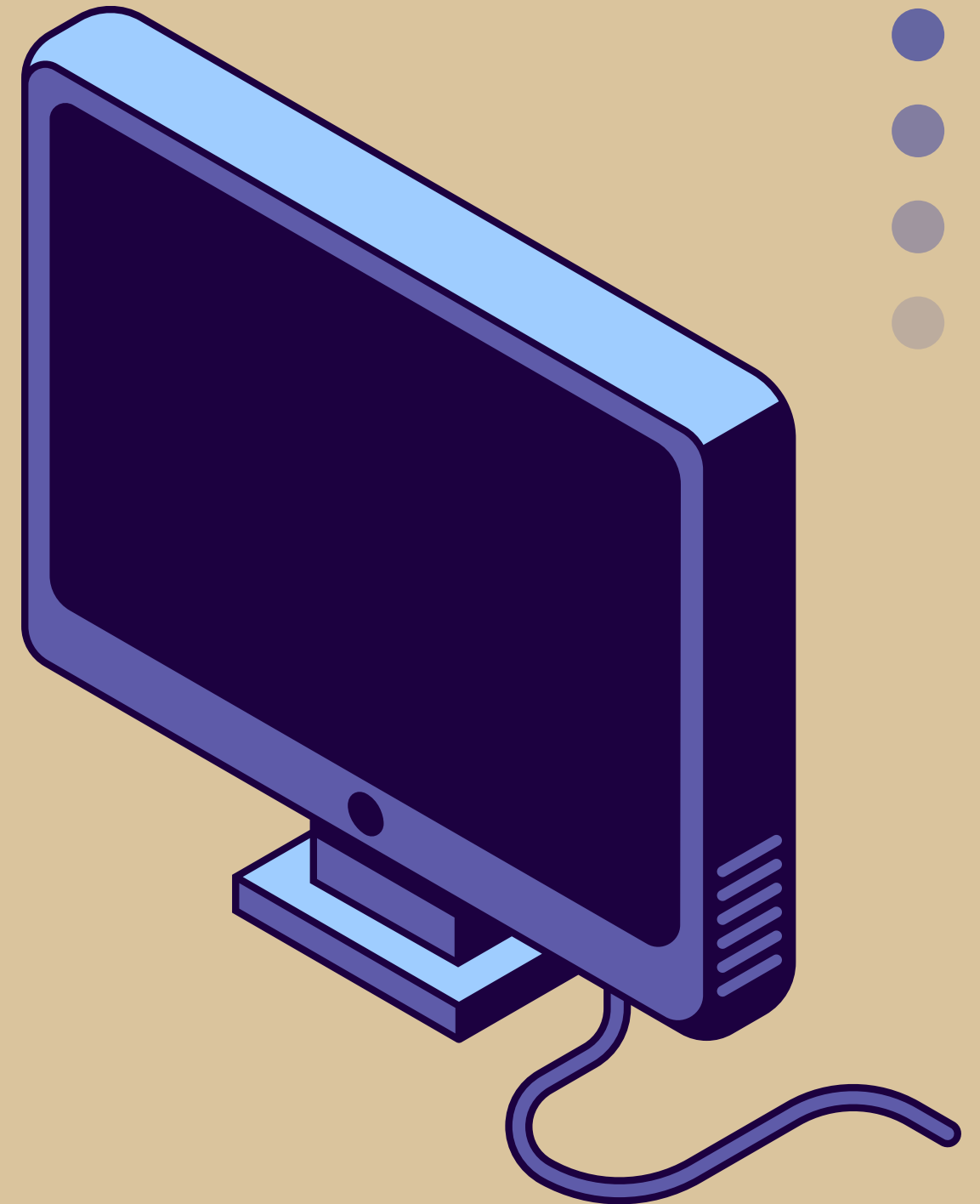




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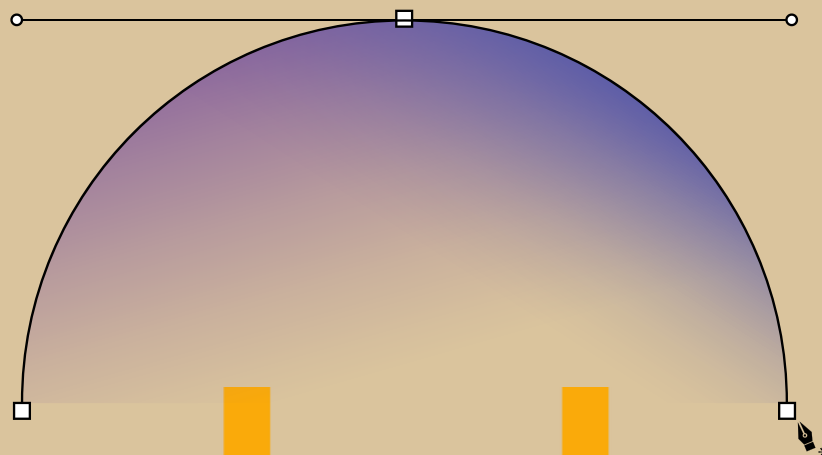
DISCOVERY AND ANALYSIS

- VirusBlokAda first detected the worm.
- Symantec and Kaspersky Lab analyzed it and revealed its complexity:
 - Over 500 KB of custom code.
 - Could update itself and hide in both Windows and PLC firmware.
 - Targeted only very specific systems to avoid detection.





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FRAMEWORK CONNECTION – NIST CSF EXAMPLE

- **Identify** : Know all assets and connected PLCs.
- **Protect** : Control USB use and apply least privilege.
- **Detect** : Monitor for abnormal PLC changes or data patterns.
- **Respond** : Isolate infected systems quickly.
- **Recover** : Rebuild clean systems and apply strict patch management.



STUXNET



THANK YOU!

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